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APPLIED RESEARCH

# Automatic Scoring System for Digitized Handwritten Answers

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**ABSTRACT** We propose a system for automatic and semiautomatic scoring of handwritten answers in Japanese, English, and mathematics, assuming the use of an electronic pen and paper. When quick scoring is prioritized, the system directly returns automatically scored results to answerers without requiring the scorers' verification. When reliable scoring is required, the system shows the results to scorers for confirmation and correction before being disclosed to the answerers. The system comprises handwriting recognizers using deep neural networks and text-based scorers to determine whether an answer is correct, incorrect, or rejected. Question-specific scoring methods and rules (rubrics) (defined by the scorers) are used to score the answers consistently. To evaluate the system, handwritten answers in the three subjects were collected from at least 50 students for all grades of elementary and junior high schools. Scoring experiments showed that the average human scoring rate (percentage of answers rejected plus those falsely negatively scored and claimed) was under 30%, indicating that approximately 70% of the scorers' workload could be reduced. The average risk scoring rate (ratio of false positives that may not be reported) was $<2\%$, except for second-grade Japanese, which requires precise character writing. Finally, we conducted feasibility experiments on the entire system, including question distribution, answer capture, automatic scoring, score correction, and score return, with participants expressing satisfaction. This study lays the groundwork for a cycle of collecting handwritten answers and improving automatic scoring in educational contexts.

**INDEX TERMS** Automatic scoring, semi-automatic scoring, handwritten answer scoring, perfect matching, similarity matching.

## I. INTRODUCTION

Recently, digitalization has been spreading rapidly into the field of education. As a result, computer-/web-based testing (CBT/WBT) is becoming popular for certification exams and small- to medium-scale examinations, such as those conducted daily.

Most CBTs and WBTs deal only with multiple-choice questions, which are easy to score automatically and suitable for touch-sensor and keyboard/mouse input on general computer terminals. However, academic achievement assessment requires the ability to think and understand in depth. Effective assessment based on multiple-choice questions depends on well-constructed and deliberate question design. An inadequate question design may negatively influence problem-solving behavior, such as encouraging learners to choose answers without thinking.

In this study, we focus on short-answer questions, particularly those requiring written answers. Short-answer questions have been used frequently in traditional (predigital

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) pen-and-paper learning environments. This format is effective for testing the understanding of a learner and reducing the cognitive load of questioning. Many of these questions exist, as these were created before digitalization. Moreover, the option to type in answers is not always available in education or examination environments and would be hard for young children. Some studies have demonstrated that handwriting is more effective than typing for retaining knowledge [1].

In examinations, handwriting is an easy or appropriate answer format in many cases, especially for mathematical and chemical formulas, drawing and graphing, and writing Kanji characters. However, the scoring of handwritten answers requires considerable time and effort. Scoring is a major burden for scorers (teachers) [2]. Recently, teacher shortages and excessive working hours have become a challenge, and teachers need to be relieved of this burden. Moreover, if it takes a long time from the time the answerer (student) answers the questions to the time the results are checked, the learning effect could be reduced owing to the fading of memory from the time of answering the questions [3]. In addition, human scoring may result in fluctuations in scoring criteria and inconsistent results among multiple scorers. Effective learning requires that scoring results be promptly presented to students so that they can immediately relearn the correct content based on an accurate understanding of their incorrect answers.

Herein, we use a tablet-type device that displays text on an electronic paper and accepts handwritten input with an electronic pen, enabling a student to write answers to questions listed in an exam paper. We chose this device because it allows the answering behavior to be recorded in the form of time-series sequences of strokes (digital ink), both online and offline recognition methods can be applied, and it is becoming less expensive and is being widely used.

However, automatic scoring of handwritten answers is still challenging. First, it is associated with misrecognition and incorrect scoring outcomes. Second, scoring must adhere to the contextual scoring criteria of the question. Third, there must be an initial milestone that demonstrates usefulness, even if it is somewhat premature. Once milestones have been established as a foundation for future development, data and experience can be accumulated to improve automated scoring.

The contributions of this study are as follows.

1. We propose a basic strategy for realizing an automatic scoring system that can be used for both automatic scoring and scoring assistance.

2. We propose a scoring strategy using perfect and similarity-based matching to reduce false results.

3. We use online and offline handwriting recognition methods to enhance reliability.

4. We formulate scoring criteria (rubrics) and propose recognition and scoring methods accordingly.

5. We realize an automatic scoring system as a first milestone that will serve as a foundation for future development.

Examinations in primary and secondary schools are mostly conducted using handwritten text; thus, we consider that the automatic scoring of handwritten text answers will be highly beneficial to teachers and students.

The paper is organized as follows. Section II summarizes previous studies. Section III clarifies technical issues and presents the basic strategies for systemization. Section IV describes the collection of handwritten answer data, and Section V reports the design and implementation of the system. Section VI evaluates the system and discusses future issues. Finally, Section VII summarizes the paper.

## II. RELATED WORKS

In this section, we review previous studies based on three paradigms, focusing on automatic scoring of descriptive answers. These studies form the basis of this research, but none are directly usable for our purpose.

### A. AUTOMATIC SCORING OF TYPED ANSWERS

The automatic scoring of descriptive answers has mostly been used for typed text. There are two main targets: automated essay scoring (AES) and short-answer scoring (SAS). Short-answer questions are scored based on content, whereas essays are scored based on rhetoric. SAS was introduced after the success of AES. Burrows et al. summarized the SAS research from 1996 to 2013, reviewing five major approaches: concept mapping, information extraction, corpus-based methods, machine learning, and assessment initiatives. They also categorized the work across six dimensions: datasets, natural language processing, model building, scoring models, model evaluation, and effectiveness validity [4].

Earlier studies on SAS used handcrafted features, which required fine-tuning on a problem-by-problem basis. Morphological analysis, synonyms, parts of speech, and keyword searches were used and evaluated on small closed datasets [5], [6].

Subsequently, the release of large open datasets and competitions accelerated research—for example, the BEETLE and SCIENTSBANK corpora [7], RIKEN-SAA [8], Kaggle's short-answer scoring competition [9], and the SemEval'13 Task 7 competition [10]. Over the past decade, deep neural networks (DNNs) and deep learning have considerably advanced the automated scoring of descriptive answers, as in other fields. These methods have outperformed models with traditional handcrafted features. In essay scoring, deep convolutional neural networks (CNNs) [11], long short-term memory (LSTM) [12] bidirectional LSTM (BLSTM) [13], and memory-augmented neural models [14] have produced promising results.

Regarding SAS, Riordan et al. [15] demonstrated improved performance over traditional non-DNN machine learning using the same basic architecture and parameter set as those used in a previous study [12]. More recently, the use of BERT [16] has led to substantial gains, often matching or surpassing human grader performance on multiple benchmark datasets

[17], [18], [19], [20]. Through pretraining, high performance has been achieved for languages other than English, such as Spanish [21] and Japanese [22], with datasets being currently available.

In typing, fill-the-gap problems are treated as solved problems because they can be easily and automatically scored with perfect matching [4]; conversely, handwritten answers are not as simple as typing due to recognition errors and the need for correct character formation.

With the availability of open datasets and following the advances in DNNs, these systems are achieving comparable accuracies to those of human graders. Depending on the learning environment, examination subjects, or target grade, however, handwriting may be more effective than keyboard input.

### B. AUTOMATIC SCORING OF HANDWRITTEN ANSWERS THROUGH OPTICAL RECOGNITION

Few studies have addressed handwritten responses. In a study on the automatic scoring of handwritten essays using optical recognition, Srihari et al. showed that the effect of misrecognition is small [23].

A team at ASSISTments, a mathematics-focused learning platform for students, organized a MathNet competition to acquire images of handwritten answers from students and used them to develop a handwriting recognition and automatic scoring system [24]. A large collection of Chinese handwritten answers provided by 2986 volunteers comprising 50000 handwritten Chinese text line images with many correction marks has been published, referred to as SCUT-EPT [25].

The development of DNNs has considerably advanced the optical recognition of handwritten answers. CNNs and RNNs were used for handwritten text recognition, combined with BLSTMs for keyword analysis in automatic answer scoring [26]. CNNs and BLSTMs were also used for essay recognition and automatic scoring [27]. BLSTMs and CNNs were used for automatic scoring of long texts and essays, achieving high performance.

Recent approaches have leveraged large language models (LLMs); for example, a previous study used ASSISTments' dataset (handwritten math expressions) and LLM-based collaborative filtering [28]. LLM-based offline character recognition has been applied to handwritten answers for English cloze questions. Its error rates have been compared with those of humans [29].

In evaluating short answers in Japanese, the scoring performance matched that of humans, even without annotation or correction of misrecognition. This was demonstrated using ensemble recognition with five representative DNN methods and automatic scoring with BERT on over 60000 answers for each of six short-answer questions ($\leq$120 characters) from trial university entrance examinations [30], [31]. However, this method is suitable for large-scale testing and requires numerous training patterns, making it unsuitable for automatic scoring in classes or schools.

This paradigm uses optical recognition of handwritten answers. The ease of scanning answer sheets and capturing their images has enabled the generation of open datasets, and recognition rates have further improved with advancements in DNNs.

### C. AUTOMATIC SCORING OF TABLET INPUT ANSWERS

Following the introduction of tablets in education, multiple workshops were organized to discuss their use [32], [33]. Many research proposals were prepared to integrate the advantages of handwriting using information technology, immediacy, shareability, use in distance education, and an understanding of overall learner trends. Despite the possibility of handwriting misrecognition, learning progress can be captured in real time, and overall statistics can be obtained quickly [34]. The scoring of handwritten words or short phrases using recognition and perfect matching has been reported and demonstrated in an English vocabulary test [35], [36].

Subsequently, proposals were prepared to cluster answers for scoring and to analyze the answering behavior recorded in digital ink [37], [38]. Because it is difficult to collect actual answers from educational settings, a composite set of correct and incorrect mathematical answers was created [39], and clustering based on handcrafted features was reported [40], [41]. Cuong et al. reported the clustering of handwritten formulas in the Competition on Recognition of Online Handwritten Mathematical Expressions dataset [42], [43] using a CNN instead of handcrafted features [44].

Recent research has shown that the accuracy of automatic grading improves when LLMs are used to correct handwritten character recognition errors [45]. A similarity-based approach that uses LLMs was also reported for the automatic grading of short handwritten answers in Japanese [46].

Tablets, which are now widely used, provide a versatile environment to answer descriptive questions and enable educators to analyze problem-solving behaviors. After their initial introduction in the education field, related research has been gaining interest recently. However, digital ink answers collected from practical educational settings remains scarce [28], [29], [47].

## III. APPROACHES TO PROBLEM SOLVING

Herein, we present a basic strategy to address the issues of handling misrecognition and incorrect scoring when implementing an automatic scoring system. The strategy involves designing a system that could be used for both automatic scoring and scoring assistance. Fig. 1 shows a schematic of this method.

### A. BASIC STRATEGIES

The automatic scoring system classifies handwritten answers into correct, incorrect, and rejected based on rubrics. Rejected leaves the decision of whether a question is correct or

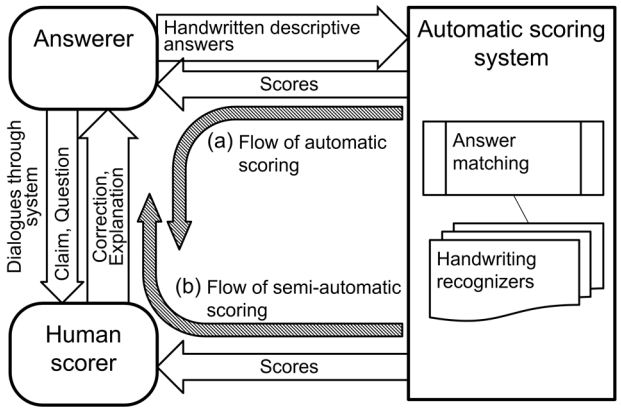



**FIGURE 1.** Scheme of the system used for both automatic and computer-assisted scoring.

incorrect to a human assessor when it is difficult to determine the answer.

Fig. 1 illustrates two flows, namely (a) and (b), after the answerer's answers are fed to the system for scoring, which represent the characteristics of the system. In Flow (a), the system functions as an automatic scorer by returning the results to the examinee before the human scorer checks or corrects them. In Flow (b), the system functions as a scoring aid by allowing the scorer to check and correct the results before returning them to the answerer.

When the answerer reviews the scoring results, they can report any errors to the scorer for correction. False negatives (*FN*), where a correct answer is judged as incorrect, should be reported. Conversely, false positives (*FP*), where an incorrect answer is judged as correct, may remain unreported. The goal is to minimize this risk. However, the addition of the confirmation process, reporting of the scoring results by the answerer, and confirmation and correction by the scorer change the paradigm of exam scoring and could increase the answerer's commitment to scoring. This approach should be applied to instructional designs and learning theories. Abuhassna et al. highlighted how instructional design models can enhance learner motivation and enable innovative approaches [48]. Reigeluth suggested that learner involvement in instructional design leads to a paradigm shift [49], aligning with our expectation.

The scoring inquiry allows the answerer to ask questions easily; additionally, even students who have not asked questions in the past do so. When responding to inquiries about scoring, except for items that should be answered individually, efficiency can be improved by answering common questions together, automatically updating the frequently asked questions, or addressing questions during review.

Flow (a) performs fully automated scoring, with an emphasis on the quick return of scoring results. This is an effective choice when the system is used for daily learning. By contrast, Flow (b) allows the scorer to mark rejected answers and check answers that are classified as correct or incorrect, thereby reducing the number of incorrect scorings and increasing the reliability of the scoring. This is essential in examinations where there is no opportunity to inquire about scoring (e.g., large-scale and certification examinations). For exams that do not require a high degree of accuracy and where the answerer can inquire about scoring, the scorer only needs to score rejected and claimed reported answers, thereby reducing the scorer's burden. Even in rejected cases, the results (correct or incorrect) associated with other answers can still be returned to the answerer immediately.

### B. EVALUATION METRICS AND INITIAL MILESTONE

We define herein three key metrics, namely the correct scoring rate, human scoring rate and risky scoring rate. The percentage of *FP* in automatic scoring is expressed as the risky scoring rate, whereas the percentage of *FN* is expressed as the claim rate. The term risky scoring rate is used because *FP*s may not be reported, while the claim-rate term is used because *FN*s will be claimed.

Answers classified as rejected, *FNs* (claims) are manually scored, and their sum denotes the human scoring rate. The rate at which automatically scored results are correct is called the correct scoring rate. This complements the sum of the risky and human scoring rates. The risky scoring and human scoring rates were set as the primary performance metrics for practical use in educational settings. These are easy to understand by the intended users (teachers and students) and serve as direct indicators of usability.

Considering the practical applications of automatic scoring systems, the risky and human scoring rates must be sufficiently low to reduce the number of incorrect scorings and rejections. We targeted a risky scoring rate at values below 2% and a human scoring rate at values below 30% because human scorers exhibit inconsistencies in correct and incorrect scorings [50], and because a reduction in the human scoring rate indicates a reduction in the scorer's labor.

It is known that the reading automation rate of handwritten addresses by early postal reading machines in the United States was approximately 50%; however, the labor savings were tens of millions of dollars [51]. This rate has been improved considerably by the progress of pattern recognition, the accumulation of data and know-how, and the advancements of DNNs. Accordingly, it is important to consider embarking on research and development endeavors on the automatic scoring of handwritten answers. The initial milestone was set up based on the history of postal sorting machines and informal interviews with teachers regarding the degree to which scoring reduction rates would be justified if automatic scoring is used.

### C. BASELINE CONSTRUCTION

The system is designed and implemented to accumulate knowledge, identify solvable problems, and collect handwritten answer data through actual use. If the system is easy to use and performs adequately well for practical applications, it can be used to collect handwritten answer data and serve

as a baseline for future enhancements. The collected data will contribute to the findings of the research studies on automatic scoring methods, and both recognition and scoring accuracy can be further improved. By repeating the cycle of data collection using the system and reflecting on the model refinement data in research, the human and risky scoring rates could be reduced further, thus potentially allowing the adoption of fully automatic scoring schemes.

## IV. COLLECTION OF HANDWRITTEN ANSWER DATA

Handwritten answer data were collected under the approval of the Ethical Review Committee of Tokyo University of Agriculture and Technology (240205-0411) with the consent of the respondents and their parents. The data were anonymized to prevent identification of the individual respondents. We collected handwritten answer data from elementary and junior high school students in Japan [52]. The students, ranging in age from 6 to 17, attended elementary school (1st to 6th grade) and junior high school (1st to 3rd).

Handwritten answer patterns were obtained from 50 elementary school students in each grade for Japanese and mathematics and from 5th and 6th grades for English as the first foreign language. The participating students comprised 145 males and 155 females. The questions were prepared using commercially available study drills and used with the permission of publishers in Japan. The numbers of questions in Japanese, mathematics, and English were 3 914, 3 490, and 318, respectively; in total, 7 722 questions were posed.

We categorized the questions into 10 types, summarized in Table 1, according to the required answer format. Japanese, English, and mathematics involve text-based answers, whereas the other types are non-text-based and are collectively categorized as line-drawing answers. Japanese answers were written vertically. Although we excluded line-drawing types, the questions used cover various types; therefore, these short-answer questions are general and appropriate for these grades. The total numbers of answers in Japanese, mathematics, and English—including blank and incorrect responses—were 97 547, 174 498, and 15 900, respectively, for an overall total of 287 945, as listed in Table 2. These numbers increase the reliability of the evaluation.

For junior high school, handwritten answers were similarly collected from 60 students per grade across all three subjects using commercially available study drills. The participating students included 81 males and 107 females. The number of questions in Japanese, mathematics, and English were 5 143, 1 566, and 4 196, respectively; in total, 10 905 questions were posed. The total numbers of answers for Japanese, mathematics, and English—including blank and incorrect responses—were 221 093, 100 206, and 262 699, respectively, for an overall total of 583 998.

Among the Japanese reading comprehension questions for junior high school students, 25 (among the total of 501) descriptive questions were answered with sentences comprising ≤60 characters, and 1 549 answers were extracted.

**TABLE 1. Ten types of answers in the adopted study drills for elementary school children.**

| Type ID | Type of answer expected |
|---|---|
| **Japanese** | Japanese text (vertical writing) |
| **English** | English text |
| **Math** | Mathematical expressions |
| **Geometry** | Completing a geometric shape by drawing additional lines |
| **Connecting** | Connecting labels with lines |
| **Locating** | Locating characters/words/phrases by drawing sidelines or underlines |
| **Selecting** | Selecting answers by gestures |
| **Filling** | Filling or painting objects such as circles, squares, and others |
| **Development** | Unfolding a 3D object to 2D |
| **Plotting** | Plotting graphs and charts |

**TABLE 2. Statistics of the handwritten answer dataset.**

| Grade | No. learners | Subject | No. of questions | No. of answers | Correct answers | Correct rate (%) |
|---|---|---|---|---|---|---|
| Elementary 1st grade | 50 | Japanese | 410 | 20 500 | 17 298 | 84.38 |
| | | Math | 464 | 23 200 | 20 973 | 90.40 |
| Elementary 2nd grade | 50 | Japanese | 563 | 28 150 | 20 792 | 73.86 |
| | | Math | 605 | 30 250 | 26 909 | 88.96 |
| Elementary 3rd grade | 50 | Japanese | 657 | 32 850 | 27 246 | 82.94 |
| | | Math | 668 | 33 400 | 29 056 | 86.99 |
| Elementary 4th grade | 50 | Japanese | 623 | 31 132 | 26 009 | 83.54 |
| | | Math | 650 | 32 498 | 28 272 | 87.00 |
| Elementary 5th grade | 50 | Japanese | 829 | 41 450 | 34 655 | 83.61 |
| | | English | 129 | 6 450 | 5 432 | 84.22 |
| | | Math | 645 | 32 250 | 28 588 | 88.64 |
| Elementary 6th grade | 50 | Japanese | 832 | 41 600 | 32 535 | 78.21 |
| | | English | 189 | 9 450 | 8 716 | 92.23 |
| | | Math | 458 | 22 900 | 21 111 | 92.19 |
| Junior 1st grade | 65 | Japanese | 1 697 | 110 305 | 86 535 | 78.45 |
| | | English | 1 335 | 86 775 | 62 587 | 72.13 |
| | | Math | 873 | 56 745 | 46 525 | 81.99 |
| Junior 2nd grade | 57 | Japanese | 1 755 | 100 035 | 77 717 | 77.69 |
| | | English | 1 431 | 81 544 | 54 480 | 66.81 |
| | | Math | 341 | 19 437 | 16 500 | 84.89 |
| Junior 3rd grade | 66 | Japanese | 1 691 | 111 596 | 84 416 | 75.64 |
| | | English | 1 430 | 94 380 | 66 167 | 70.11 |
| | | Math | 352 | 23 232 | 18 604 | 80.08 |

Each answer was labeled as correct or incorrect by a qualified teacher. The percentage of correct answers ranged from 66% to 92%. Table 2 presents the statistics on the number of questions and answers. Each subject was further divided into two categories for junior high school students: reading comprehension and Kanji words for Japanese; regular math and calculation problems for mathematics; and regular English problems and English vocabulary for English.

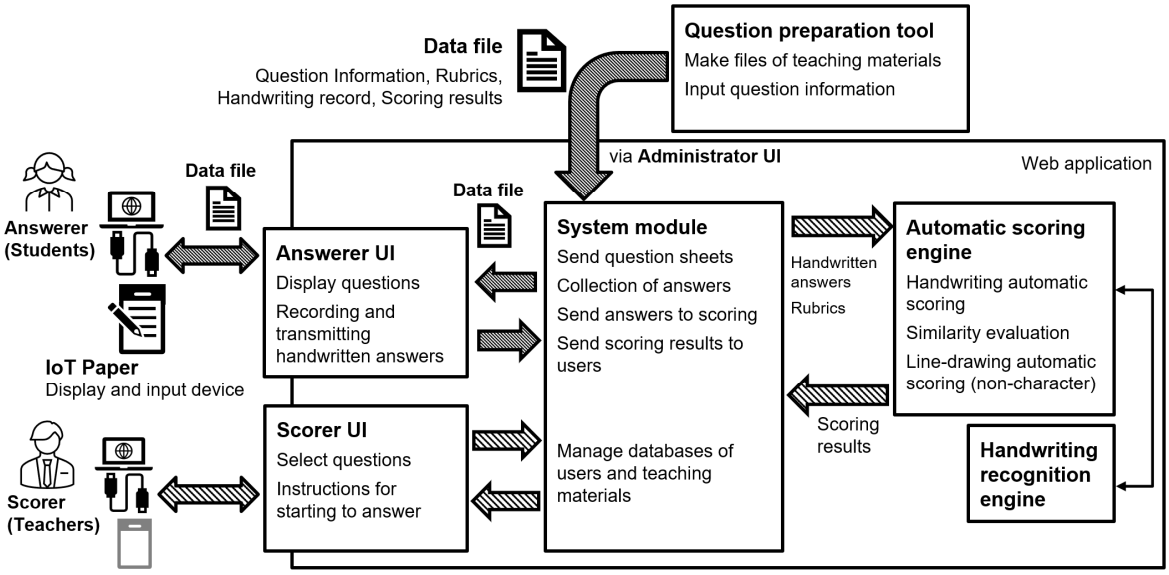



**FIGURE 2.** Configuration of the automatic scoring system.

From these answers, 10% were randomly selected as the validation set and used to tune the threshold. The remainder is used as the test set. The optimized thresholds are presented in Section VI-A.

## V. BASIC DESIGN AND IMPLEMENTATION

The system displays questions on an electronic paper, records the answers written using an electronic pen, recognizes the answers, and scores them according to a rubric by comparing the top recognition candidates with the expected correct answers. In this study, we focused on answers to questions across three categories: Japanese, English, and mathematics.

Fig. 2 shows the system configuration. As shown in Fig. 3, the system is used through a web application using a browser on each user's computer terminal connected to an electronic paper and pen. We selected a web application format because it reduces operating system (OS) and device dependence. The system includes an automatic scoring function on the server side; therefore, even if the user's terminal is not sufficiently powerful, the automatic scoring function can still be used. It also enables easy collection of answer data and facilitates updates to the functionality and performance of the automatic scoring method.

Details of each module in the configuration diagram are described in the following subsections.

### A. IOT PAPER—DISPLAY/INPUT DEVICE

For this system, we prepared an electronic paper and pen named IoT Paper, a device mainly used for displaying questions and inputting handwritten answers.

An electronic paper display (EPD) is used in the display part of the IoT Paper, which uses pigments instead of

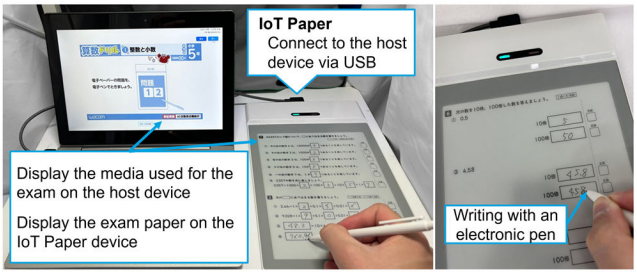
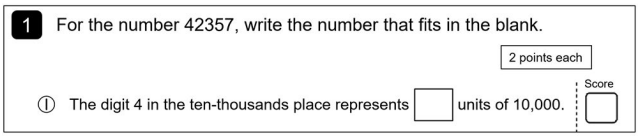



**FIGURE 3.** Photographs demonstrating the use of the writing device.

light-emitting devices to create a display similar to printed paper. In monochrome EPDs, negatively charged white and positively charged black pigments are enclosed in microcapsules; the application of voltages causes the pigments to move electrophoretically, resulting in an arbitrary screen display. The EPD has a display area of 207.3 mm × 154.9 mm and a display resolution of 1870 × 1402 pixels.

An electronic pen is used for input. The sampling rate, coordinate accuracy, and maximum reading height of the electronic pen over the surface are 480 Hz, 0.01 mm, and 10 mm, respectively. The X/Y coordinates, height, writing pressure, and pen tilt in the X/Y direction (pen trajectory values) are acquired in a single scan and recorded as a time series in a file with a timestamp.

The IoT Paper measures 297 mm × 182 mm; it is 8 mm thick at the top (where the controller board is housed) and 3 mm thick at the display and handwriting area, which is thinned to facilitate writing on a desk. The IoT Paper is not equipped with a touch sensor and has no physical buttons; however, it includes a light-emitting diode indicator to show whether it is writable. Thus, users can concentrate on answering by handwriting. The IoT Paper does not have a battery and




(a) A fill-in-the-blank question of mathematics.

```
<annotation type="QARA">
  <input xml:id="1" x="645" y="205" width="140" height="115"/>
  <question xml:id="q1" category="math">
    <answer xml:id="aw1" inputIdRef="#1" recognizer="math"/>
  </question>
  <rubrics questionIdRef="#q1" min="0" max="1">
    <criterion xml:id="cr1" answerIdRef="#aw1"
               expectedValue="4" algorithm="matching"
               min="0" max="1"/>
  </rubrics>
</annotation>
```

(b) IDPML format for the question shown above. The <input> element provides the coordinates of the answer field, while the <criterion> element represents an example of correct answer and corresponding scoring method.

**FIGURE 4. Example of a question that appears on the IoT Paper device, along with its metadata in IDPML format.**

is connected to a computer terminal via a universal serial bus (USB).

### B. IDPML – DATA FILE FORMAT

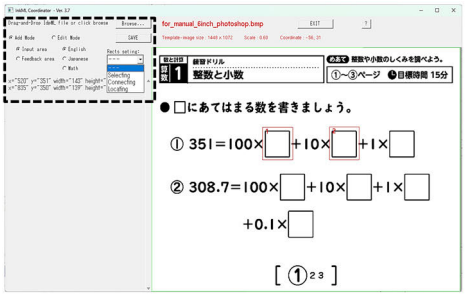
Questions and handwritten answers are exchanged as files between the automatic scoring system and IoT Paper. To describe the data in these files, we defined IDPML as a special file format for expressing information regarding questions and records of handwritten answers. This format conforms to eXtensible Markup Language. One file corresponds to one sheet of paper, and the question and handwritten data are stored in one file as one sheet of question paper.

The pen trajectory values are recorded as handwritten data using the Ink Markup Language (InkML) format, which is standardized by the World Wide Web Consortium [53]. The values scanned by the electronic pen's sensor are recorded in a time series. By rendering the coordinates and pen pressure values included in the data, handwritten content can be easily reconstructed as an image. Unlike optical character recognition, this method has the advantage of isolating the handwritten content for data processing without the need to remove the printed background or extract the handwriting separately.

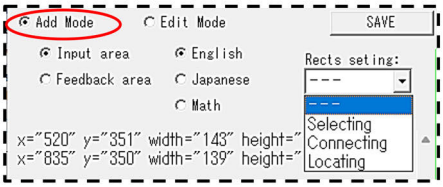
Information regarding the questions is defined separately from the InkML data as a unique element. The question paper displayed on the IoT Paper is a monochrome bitmap (BMP) image file, and the image data are stored as a string in base64 format. Additionally, information for automatic scoring (scoring criteria/rubric) is included.

### C. QUESTION INFORMATION DESCRIPTION AND QUESTION PREPARATION TOOL

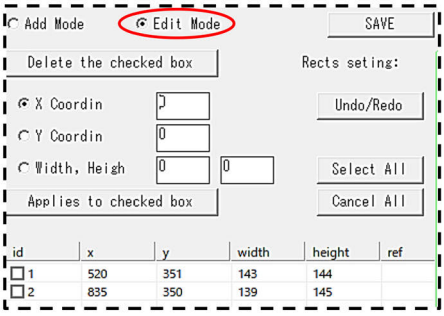
Among the data related to the questions, the coordinates of the answer boxes and the rubric are necessary for automatic scoring. Coordinates are required to extract the handwritten




(a) Register the exam paper image.




(b) The add mode is used to declare question information and insert answer areas.




(c) The edit mode is used to adjust the existed answer areas.

**FIGURE 5. Question preparation tool.**

answers to be scored from the entire answer sheet. For each question, the rubric specifies the recognition method to be used, method of automatic scoring, additional information to be used for recognition and scoring, and scoring criteria. As each question differs in what it asks and the type of response it values, it is essential to define clearly the scoring criteria. In addition, the automatic scoring system operates based on rubric specifications to ensure uniform scoring performance. This is important for the answerer's user experience. Fig. 4 shows an example of a question and its information description.

We prepared the question-preparation tool shown in Fig. 5 to set the rubric in the IDPML file. Once the question paper image is developed, the answer box coordinates and question type are specified using this tool. This tool makes it easy for question builders to prepare question data for learning, even if they are not familiar with the question format. Furthermore, as more knowledge is accumulated regarding question types and appropriate rubrics, it becomes possible to implement a function that allows users to prepare questions by selecting registered subquestions or to help users automatically select questions for a specific unit of study.

The prepared question files are registered with the system as a single teaching material dataset by combining multiple

IDPML (equivalent to a multipage exam) and media files. Registration is performed in advance via the administrator's user interface (UI) before teachers and students use the system for teaching and learning.

### D. HANDWRITING RECOGNITION ENGINE

For handwritten answer recognition, we used recognition engines for handwritten Japanese, English, and mathematical equations, corresponding to the three question types, namely Japanese, English, and mathematics, all of which involved answers that used handwritten characters. For each recognition engine, we used an online recognition engine that utilized time-series data from handwriting coordinates and an offline recognition engine that used rendered image data of the handwriting. It is easy to render handwriting image characters for the offline recognition method from the time-series coordinate data of handwritten strokes. The online recognition method is effective for verifying whether the stroke order and structure are correct, whereas the offline method can evaluate whether the glyphs are correctly written, even if the strokes are out of order or rewritten. Together, these methods provide a stable recognition approach for stylistic variations. The recognition functions to be used are specified in the rubric.

Because the system's target audience includes younger children, handwriting style deviations may be greater than what conventional recognition engines typically accommodate. Furthermore, if a misspelled or non-existent character is incorrectly recognized as correct, it becomes unsuitable for scoring. Therefore, a context-aware recognition process cannot be used.

As recognition engines for the online and offline recognition methods of each language, we adopted a commercialized recognition engine developed by iLabo, based on research studies on the recognition of handwritten Japanese [54], [55], English [56], and mathematical formulas [57], [58], using DNNs. None of the handwritten answer patterns collected in this study were used for training the recognition engines.

### E. AUTOMATIC SCORING ENGINE

The automatic scoring engine received recognition results from the handwriting recognition engine for handwritten answers in Japanese, English, and mathematical formulas. Thereafter, it scored them by referring to examples of correct answers and rubrics without correcting any errors or annotating them. There are two matching methods: perfect matching and Levenstein edit distance similarity (text-based similarity). The matching method is specified in the rubric.

Perfect matching is used based on whether any of the candidates recognized by the engine match one of the correct answer examples. In the case of $M$ recognized candidates and $N$ example correct answers, the score ($\text{score}_{\text{PM}}$) is expressed according to the following formula based on the edit distance, EditDistance(), between the $j$-th recognized candidate ($s^j_{\text{rc}}$)

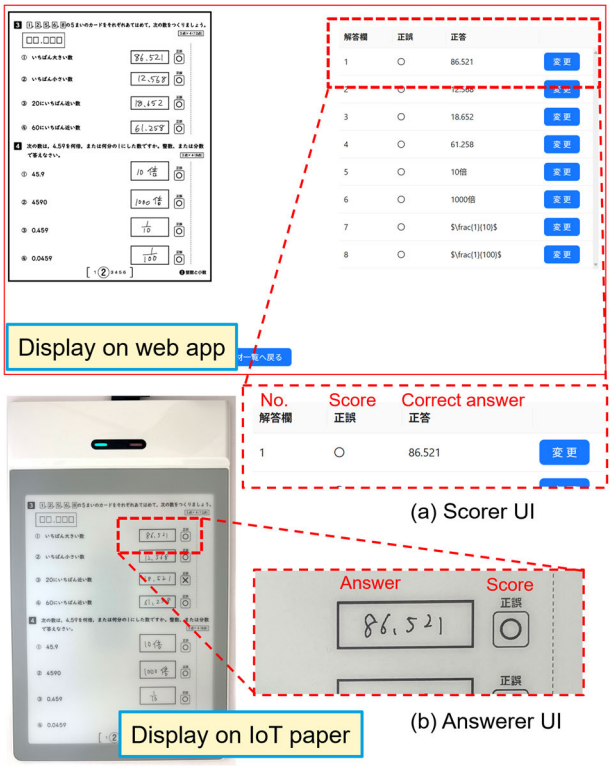



**FIGURE 6.** Confirmation of scored results by teacher and student.

and $i$-th exemplar answer, ($s^i_{\text{ca}}$):

$$\text{score}_{\text{PM}} = \max_{\substack{j\in\{1,\ldots,M\}\\ i\in\{1,\ldots,N\}}} \left( \begin{cases} 1 \text{ if EditDistance } \left(s^j_{\text{rc}}, s^i_{\text{ca}}\right) = 0 \\ 0 \text{ if EditDistance } \left(s^j_{\text{rc}}, s^i_{\text{ca}}\right) \neq 0 \end{cases} \right) \tag{1}$$

Similarity judgments aim to reduce the numbers of $FP$s and $FN$s by rejecting some scores and allowing them to be scored by humans. Similarity is expressed by the following equation, using the edit distance of the text between the recognized candidates and example correct answer:

$$\text{seq_ sim}\left(s^j_{\text{rc}}, s^i_{\text{ca}}\right) = 1 - \frac{\text{EditDistance}\left(s^j_{\text{rc}}, s^i_{\text{ca}}\right)}{\max\left(\text{len}\left(s^j_{\text{rc}}\right), \text{len}\left(s^i_{\text{ca}}\right)\right)} \tag{2}$$

$$\text{score}_{\text{TS}} = \max_{j\in\{1,\ldots,M\}, i\in\{1,\ldots,N\}} \left\{\text{seq_sim}\left(s^j_{rc}, s^i_{ca}\right)\right\} \tag{3}$$

Among the recognized candidates, the edit distance closest to the model answer is adopted as the similarity outcome ($\text{score}_{\text{TS}}$). Notably, len() returns the number of characters in the string. Two threshold values are set for this similarity outcome. The similarity outcome ($\text{score}_{\text{TS}}$) is classified as a correct answer if it exceeds the high threshold value $\delta_{\text{high}}$, and it is classified as an incorrect answer if it is lower than the low threshold, $\delta_{\text{low}}$. Conversely, the answer is rejected if neither of these assumptions are valid (that is the similarity outcome is between the two thresholds).

### F. UI OF THE ANSWERER

The answerer (student) accesses the system using a computer terminal connected to the IoT Paper. The questions specified by the scorer (teacher) are listed as answerable materials, and the student uses the IoT Paper to answer the questions.

While answering a question, a question paper is displayed on the IoT Paper, and the answer is written on it. Simultaneously, the browser can display the media (images, audio, and videos). This makes it possible to play audio during a listening test. Because IoT Paper does not have any input method other than handwritten input, the browser can be used to select the material, start answering questions, move pages, and finish answering questions.

When automatic scoring is used, the results can be checked immediately after completing the exam. When used as a scoring aid, the student must wait for the scorer to review the rejected answers and confirm or correct the automatic scoring results. The scorer must check and correct the results of the automatic scoring and wait for the scorer to score the rejected answers. The scoring outcomes can be viewed on the IoT Paper by displaying the answer sheet with marks in the correct and incorrect columns (Fig. 6). If there are any errors in the scoring results, the student should report them to the teacher.

### G. UI OF THE SCORER

The scorer accesses the system, searches for preregistered materials, specifies the materials to be used for the upcoming test, and distributes them to the answerer. After distribution, the IoT Paper connection status and time remaining for each respondent can be checked, and the progress of the test can be monitored.

When used for automatic scoring, the scorer waits for the answerer's claim before reviewing and correcting the score. However, the scoring results can be viewed at any time after the answerer has completed the questions.

When used as a scoring aid, the scorer reviews the automatic scoring results immediately after the exam is completed and corrects them if necessary. Additionally, a scorer can score the rejected answers, complete the scoring of the entire answers, and allow the answerer to check the scoring results.

In addition to viewing the answers of each student, the scorer UI displays a list of all the students' answers for each question and allows the scorer to check the results of automatic scoring and make corrections. The scorer can operate the system with all students' answers displayed side by side and categorized as correct or incorrect, allowing for quick checking of the scoring results (Fig. 7). The list view makes efficient scoring, particularly when the system is used as a scoring aid.

The results of automatic scoring and corrections are recorded in the file. The confirmation of scoring results and the recording of corrections and their histories ensure the transparency of scoring, encourage review immediately after the exam, and increase respondents' active participation in the exam. By involving the answerers in score correction, we could promote communication between the answerer and scorer (student and teacher) to solve the problem.

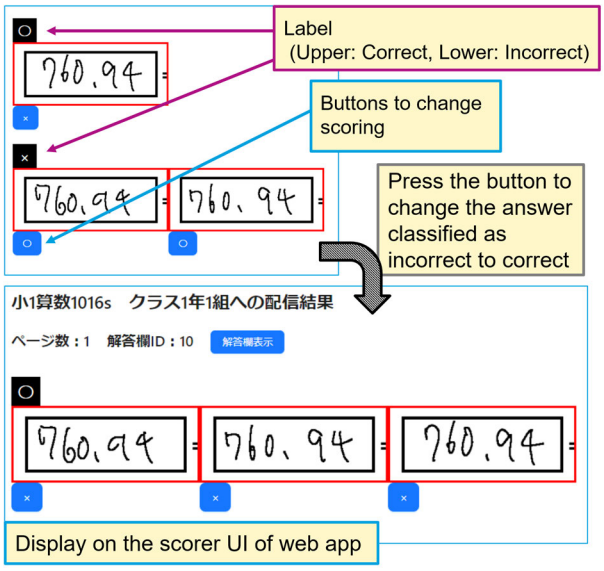



**FIGURE 7.** Scorer UI used for correcting scores based on a list of answers.

### H. SYSTEM IMPLEMENTATION

This system is implemented as a Web application and has a main system module. The system module manages the database of teaching materials and user information, providing a user interface for each user to access data (distribution and retrieval of questions, checking scoring, and management of user information) and transmission and reception of answer data and scoring results to and from the automatic scoring engine.

The system provides the following functions for three types of users: administrators, teachers (scorers), and students (answerers).

- Administrators: manage and add school and teacher information, activate or deactivate account access, manage and add student information, re-issue passwords, add and deactivate materials, check automatic scoring errors, and check user status.
- Teachers: submit tests, manage answers and scoring results, score rejected answers, and correct errors in scoring.
- Students (must be connected to IoT Paper): answering questions (taking tests or learning through drills) and checking scoring results.

The target operating systems for the web application are Windows and ChromeOS, and operation has been confirmed using Edge and Chrome browsers. The application was developed using Python, Go, and TypeScript and deployed on the AWS.

The IoT Paper is recognized as an MTP device by the computer terminal, and its basic operation is based on sending and receiving files to and from the Web app via the WebUSB

**TABLE 3. Finetuned values of the two thresholds for the text-based similarity method.**

| Answer type | Subject/Category | Grade | Thresholds | |
|---|---|---|---|---|
| | | | $\delta_{high}$ | $\delta_{low}$ |
| Japanese single character | Japanese | Elementary grades | .90 | .50 |
| | Reading Comprehension | Junior high school grades | .40 | .20 |
| | Kanji Vocabulary | | .90 | .50 |
| Japanese text (multiple characters) | Japanese | Elementary grades | .90 | .80 |
| | Reading Comprehension | Junior high school grades | .70 | .70 |
| | Kanji Vocabulary | | .90 | .80 |
| English | All subjects | All grades | .95 | .95 |
| Math | All subjects | All grades | .50 | .50 |

application programming interface. That is, the computer terminal (and thus the web app) can place and retrieve files from the IoT Paper as a storage device at any time. When IoT Paper receives a file, it interprets the contents of the file and performs actions according to the contents. IoT Paper records the contents of the file, and the web app retrieves the file for automatic scoring.

The system module extracts the handwritten answer data and rubric from the file and passes them to the automatic scoring engine. The automatic scoring engine uses a handwriting recognition engine to output the results of automatic scoring based on the rubric.

Although the rubric can specify how to select and combine online and offline recognition for automatic scoring, we simply combined the top candidates for online or offline recognition. In addition, while the rubric allows detailed specification of scoring criteria for each question, a uniform default was initially applied to all questions. For the Japanese course on writing Kanji characters neatly (calligraphy), we also implemented a calligraphic function that evaluated key components of proper Kanji writing (balance, stroke order, and stroke ending features: stopping, flicking, and sweeping); however, this function was not used at this time.

## VI. PRELIMINARY SYSTEM EVALUATION

Herein, we present a preliminary evaluation of the system.

### A. QUANTITATIVE EVALUATION OF AUTOMATIC SCORING PERFORMANCE

Two automatic scorers, perfect matching (PM) and text-based similarity (TbS), were applied to all the answers provided to the Japanese, English, and mathematics questions obtained from the study drills for elementary (E1 to E6) and junior high schools (J1 to J3). Note that the two thresholds $\delta_{high}$ and $\delta_{low}$ of the text-based similarity method were fine-tuned based on the validation set to satisfy (or to be as close as possible

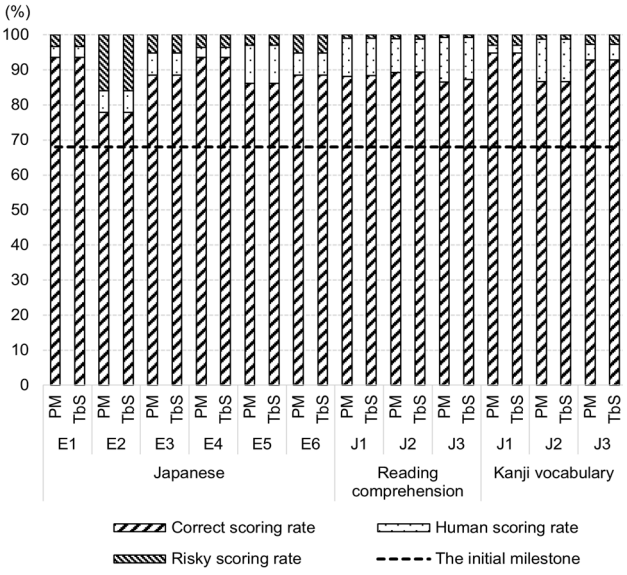



**FIGURE 8. Performance outcomes (%) of automatic scorers on the handwritten single-character answers of Japanese text.**

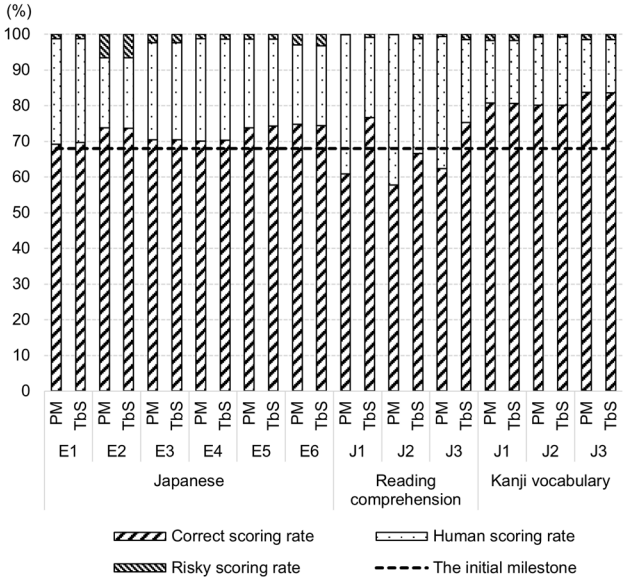



**FIGURE 9. Performance outcomes (%) of automatic scorers on the handwritten answers of Japanese text (multiple characters).**

to) the initial milestone, where the human and risky scoring rates were set at values less than 30% and 2%, respectively. Because of the different characteristics of the answer types as well as subjects and categories, the fine-tuned thresholds are listed in Table 3.

The performance outcomes of the automatic scorers are shown in Figs. 8–11 in conjunction with the initial milestone (dashed line) of the correct scoring rate at 68% that is the complement of the human scoring rate of 30% and risky scoring rate of 2%. The detailed results are presented in Tables 4–7. These tables list average performance outcomes

**TABLE 4.** Performance outcomes (Mean ± Standard error%) of automatic scorers on the handwritten answers of Japanese text comprising single characters.

| Subject/ Category | Grade | Number of test samples | Scoring method | Correct scoring rate | Human scoring rate | Risky scoring rate |
|---|---|---|---|---|---|---|
| Japanese | Elementary 1st grade | 7 968 | PM | 93.62 ± .27 | 3.07 ± .19 | 3.30 ± .20 |
| | | | TbS | 93.62 ± .27 | 3.07 ± .19 | 3.30 ± .20 |
| | Elementary 2nd grade | 11 924 | PM | 77.83 ± .38 | 6.21 ± .22 | 15.96 ± .34 |
| | | | TbS | 77.83 ± .38 | 6.21 ± .22 | 15.96 ± .34 |
| | Elementary 3rd grade | 13 634 | PM | 88.56 ± .27 | 6.21 ± .21 | 5.24 ± .19 |
| | | | TbS | 88.56 ± .27 | 6.21 ± .21 | 5.24 ± .19 |
| | Elementary 4th grade | 11 741 | PM | 93.61 ± .23 | 2.78 ± .15 | 3.61 ± .17 |
| | | | TbS | 93.61 ± .23 | 2.78 ± .15 | 3.61 ± .17 |
| | Elementary 5th grade | 17 009 | PM | 86.13 ± .27 | 10.94 ± .24 | 2.93 ± .13 |
| | | | TbS | 86.13 ± .27 | 10.94 ± .24 | 2.93 ± .13 |
| | Elementary 6th grade | 16 739 | PM | 88.51 ± .25 | 6.32 ± .19 | 5.17 ± .17 |
| | | | TbS | 88.51 ± .25 | 6.32 ± .19 | 5.17 ± .17 |
| Reading comprehension | Junior 1st grade | 5 907 | PM | 88.15 ± .42 | 10.85 ± .40 | 1.00 ± .13 |
| | | | TbS | 88.39 ± .42 | 10.60 ± .40 | 1.02 ± .13 |
| | Junior 2nd grade | 6 103 | PM | 89.28 ± .40 | 9.57 ± .38 | 1.15 ± .14 |
| | | | TbS | 89.40 ± .39 | 9.36 ± .37 | 1.25 ± .14 |
| | Junior 3rd grade | 7 418 | PM | 86.52 ± .40 | 12.78 ± .39 | 0.70 ± .10 |
| | | | TbS | 87.30 ± .39 | 11.96 ± .38 | 0.74 ± .10 |
| Kanji vocabulary | Junior 1st grade | 23 457 | PM | 94.86 ± .14 | 2.16 ± .09 | 2.98 ± .11 |
| | | | TbS | 94.86 ± .14 | 2.16 ± .09 | 2.98 ± .11 |
| | Junior 2nd grade | 20 724 | PM | 86.71 ± .24 | 12.03 ± .23 | 1.26 ± .08 |
| | | | TbS | 86.71 ± .24 | 12.03 ± .23 | 1.26 ± .08 |
| | Junior 3rd grade | 21 739 | PM | 92.82 ± .18 | 4.41 ± .14 | 2.77 ± .11 |
| | | | TbS | 92.82 ± .18 | 4.41 ± .14 | 2.77 ± .11 |

PM: Perfect matching, TbS: Text-based similarity

**TABLE 5.** Performance outcomes (Mean ± Standard error%) of automatic scorers on the handwritten answers of japanese text comprising multiple characters.

| Subject/ Category | Grade | Number of test samples | Scoring method | Correct scoring rate | Human scoring rate | Risky scoring rate |
|---|---|---|---|---|---|---|
| Japanese | Elementary 1st grade | 10 270 | PM | 69.14 ± .46 | 29.67 ± .45 | 1.19 ± .11 |
| | | | TbS | 69.64 ± .45 | 29.16 ± .45 | 1.20 ± .11 |
| | Elementary 2nd grade | 13 319 | PM | 73.86 ± .38 | 19.60 ± .34 | 6.54 ± .21 |
| | | | TbS | 73.72 ± .38 | 19.67 ± .34 | 6.61 ± .22 |
| | Elementary 3rd grade | 15 929 | PM | 70.44 ± .36 | 27.24 ± .35 | 2.32 ± .12 |
| | | | TbS | 70.48 ± .36 | 27.18 ± .35 | 2.34 ± .12 |
| | Elementary 4th grade | 16 275 | PM | 70.02 ± .36 | 28.69 ± .35 | 1.28 ± .09 |
| | | | TbS | 70.29 ± .36 | 28.39 ± .35 | 1.32 ± .09 |
| | Elementary 5th grade | 20 294 | PM | 73.78 ± .31 | 24.86 ± .30 | 1.36 ± .08 |
| | | | TbS | 74.24 ± .31 | 24.37 ± .30 | 1.39 ± .08 |
| | Elementary 6th grade | 20 654 | PM | 74.81 ± .30 | 22.26 ± .29 | 2.93 ± .12 |
| | | | TbS | 74.42 ± .30 | 22.41 ± .29 | 3.17 ± .12 |
| Reading comprehension | Junior 1st grade | 5 907 | PM | 60.91 ± .63 | 38.95 ± .63 | 0.14 ± .05 |
| | | | TbS | 76.65 ± .55 | 22.53 ± .54 | 0.81 ± .12 |
| | Junior 2nd grade | 6 308 | PM | 57.77 ± .62 | 42.14 ± .62 | 0.10 ± .04 |
| | | | TbS | 66.61 ± .59 | 32.15 ± .59 | 1.24 ± .14 |
| | Junior 3rd grade | 5 400 | PM | 62.41 ± .66 | 37.00 ± .66 | 0.59 ± .10 |
| | | | TbS | 75.26 ± .59 | 23.28 ± .58 | 1.46 ± .16 |
| Kanji vocabulary | Junior 1st grade | 63 998 | PM | 80.74 ± .16 | 17.58 ± .15 | 1.69 ± .05 |
| | | | TbS | 80.64 ± .16 | 17.67 ± .15 | 1.69 ± .05 |
| | Junior 2nd grade | 56 890 | PM | 80.16 ± .17 | 19.08 ± .16 | 0.76 ± .04 |
| | | | TbS | 80.11 ± .17 | 19.13 ± .16 | 0.76 ± .04 |
| | Junior 3rd grade | 65 873 | PM | 83.70 ± .14 | 14.79 ± .14 | 1.50 ± .05 |
| | | | TbS | 83.64 ± .14 | 14.86 ± .14 | 1.50 ± .05 |

without standard errors because they are small, as explained below.

Fig. 8 and Table 4 show the results for single-character answers; Fig. 9 and Table 5 show those for text strings, using PM and TbS for Japanese answers. Overall, the automatic scorers satisfied the initial milestone for the Japanese questions except for some grades.

Regarding Japanese answers, the automatic scoring results using PM and similarity were comparable for single-character questions. The human scoring rate remained below

**TABLE 6.** Performance outcomes (Mean ± Standard error%) of automatic scorers on handwritten english answers.

| Subject/ Category | Grade | Number of test samples | Scoring method | Correct scoring rate | Human scoring rate | Risky scoring rate |
|---|---|---|---|---|---|---|
| English | Elementary 5th grade | 5 804 | PM | 91.02 ± .38 | 8.01 ± .36 | 0.96 ± .13 |
| | | | TbS | 91.14 ± .37 | 7.46 ± .34 | 1.40 ± .15 |
| | Elementary 6th grade | 8 504 | PM | 95.57 ± .22 | 3.08 ± .19 | 1.35 ± .13 |
| | | | TbS | 95.57 ± .22 | 3.08 ± .19 | 1.35 ± .13 |
| | Junior 1st grade | 49 139 | PM | 85.45 ± .16 | 11.34 ± .14 | 3.22 ± .08 |
| | | | TbS | 86.29 ± .16 | 9.59 ± .13 | 4.12 ± .09 |
| | Junior 2nd grade | 37 793 | PM | 88.78 ± .16 | 7.94 ± .14 | 3.29 ± .09 |
| | | | TbS | 88.89 ± .16 | 6.61 ± .13 | 4.51 ± .11 |
| | Junior 3rd grade | 50 429 | PM | 90.61 ± .13 | 8.79 ± .13 | 0.60 ± .03 |
| | | | TbS | 92.13 ± .12 | 6.93 ± .11 | 0.94 ± .04 |
| English vocabulary | Junior 1st grade | 28 956 | PM | 87.61 ± .19 | 11.75 ± .19 | 0.65 ± .05 |
| | | | TbS | 88.02 ± .19 | 11.23 ± .19 | 0.74 ± .05 |
| | Junior 2nd grade | 35 594 | PM | 82.72 ± .20 | 13.49 ± .18 | 3.79 ± .10 |
| | | | TbS | 82.97 ± .20 | 12.93 ± .18 | 4.10 ± .11 |
| | Junior 3rd grade | 34 510 | PM | 85.99 ± .19 | 13.48 ± .18 | 0.53 ± .04 |
| | | | TbS | 86.90 ± .18 | 12.50 ± .18 | 0.60 ± .04 |

**TABLE 7.** Performance outcomes (Mean ± Standard error%) of automatic scorers on handwritten mathematics answers.

| Subject/ Category | Grade | Number of test samples | Scoring method | Correct scoring rate | Human scoring rate | Risky scoring rate |
|---|---|---|---|---|---|---|
| Math | Elementary 1st grade | 20 879 | PM | 44.65 ± .34 | 54.94 ± .34 | 0.41 ± .04 |
| | | | TbS | 82.16 ± .26 | 16.27 ± .26 | 1.57 ± .09 |
| | Elementary 2nd grade | 27 224 | PM | 30.52 ± .28 | 69.42 ± .28 | 0.06 ± .02 |
| | | | TbS | 83.28 ± .23 | 15.08 ± .22 | 1.65 ± .08 |
| | Elementary 3rd grade | 30 059 | PM | 46.79 ± .29 | 52.89 ± .29 | 0.32 ± .03 |
| | | | TbS | 92.29 ± .15 | 5.99 ± .14 | 1.72 ± .07 |
| | Elementary 4th grade | 29 247 | PM | 44.28 ± .29 | 55.33 ± .29 | 0.39 ± .04 |
| | | | TbS | 88.57 ± .19 | 9.96 ± .18 | 1.48 ± .07 |
| | Elementary 5th grade | 29 024 | PM | 46.40 ± .29 | 53.33 ± .29 | 0.27 ± .03 |
| | | | TbS | 89.11 ± .18 | 9.92 ± .18 | 0.97 ± .06 |
| | Elementary 6th grade | 20 609 | PM | 67.66 ± .33 | 31.99 ± .32 | 0.35 ± .04 |
| | | | TbS | 94.31 ± .16 | 4.97 ± .15 | 0.72 ± .06 |
| | Junior 1st grade | 20 474 | PM | 37.33 ± .34 | 62.63 ± .34 | 0.04 ± .01 |
| | | | TbS | 84.50 ± .25 | 14.99 ± .25 | 0.50 ± .05 |
| | Junior 2nd grade | 7 437 | PM | 29.52 ± .53 | 70.47 ± .53 | 0.01 ± .01 |
| | | | TbS | 81.58 ± .45 | 17.79 ± .44 | 0.63 ± .09 |
| | Junior 3rd grade | 7 958 | PM | 30.21 ± .51 | 68.84 ± .52 | 0.96 ± .11 |
| | | | TbS | 70.96 ± .51 | 26.77 ± .50 | 2.27 ± .17 |
| Calculation | Junior 1st grade | 30 594 | PM | 24.98 ± .25 | 74.99 ± .25 | 0.03 ± .01 |
| | | | TbS | 81.04 ± .22 | 18.27 ± .22 | 0.69 ± .05 |
| | Junior 2nd grade | 10 053 | PM | 25.19 ± .43 | 74.79 ± .43 | 0.02 ± .01 |
| | | | TbS | 80.48 ± .40 | 18.99 ± .39 | 0.53 ± .07 |
| | Junior 3rd grade | 12 948 | PM | 35.54 ± .42 | 64.41 ± .42 | 0.05 ± .02 |
| | | | TbS | 79.05 ± .36 | 19.69 ± .35 | 1.25 ± .10 |

15% in both methods, with a risky scoring rate under 2% for junior high school reading comprehension. However, the risky scoring rate was higher for Kanji vocabulary among junior high school students and for second-grade elementary students, likely due to stricter glyph evaluation by certified teachers. This issue is discussed in the next section.

For Japanese answer questions using text comprising two or more characters (multiple characters), the human scoring rate remained below 30% with PM, except for reading comprehension and first-grade students. The risky scoring rate remained below 2% except in the second, third, and sixth-grade elementary school groups. The similarity scores for elementary grades and Kanji vocabulary were stable across threshold values, but adjusting the threshold improved the human scoring rate in reading comprehension questions.

Fig. 10 and Table 6 illustrate the results for English in which either PM or similarity scorers satisfied the initial milestone across all grades. These scorers exhibited similar performances, and the human scoring rate for both methods was low and varied from 3% to 14%. The risky scoring rate was lower than 2% for elementary school students and less than 1% for third-year junior high school students; however, it reached approximately 3% for first- and second-year junior high school students. Therefore, it is necessary to consider methods to reduce the risky scoring rate for a particular grade.

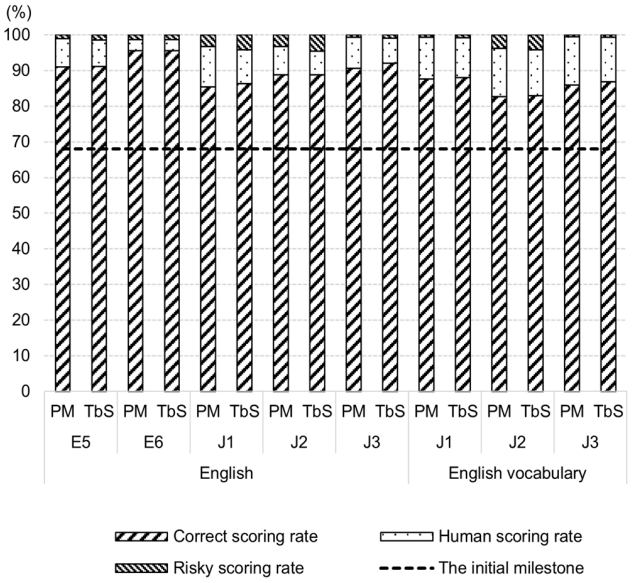



**FIGURE 10.** Performance outcomes (%) of automatic scorers on handwritten English answers.

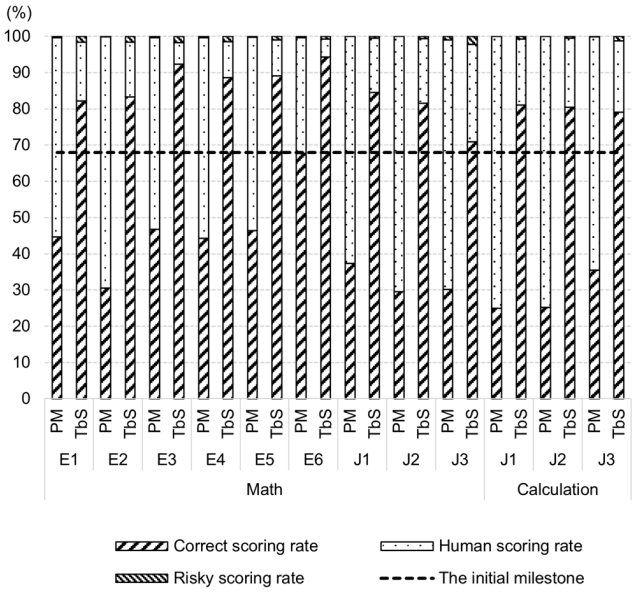



**FIGURE 11.** Performance outcomes (%) of automatic scorers on handwritten mathematics answers.

The results for mathematics are presented in Fig. 11 and Table 7. With PM, the human scoring rate exceeded 30%. However, the use of similarity scoring considerably improved performance, achieving risky and human scoring rates below 2% and 30%, respectively, except for the risky scoring rate in third-year junior high school mathematics. Thus, the similarity scorer achieved the initial milestone at all grades.

The performance outcomes of all the studied subjects (means ± standard errors %) are summarized as follows. When an appropriate scoring method is selected, the human and risky scoring rates for Japanese were 15.46 ± 0.02% and 2.49 ± 0.02%, respectively; for English, the corresponding rates were 10.60 ± 0.02% and 2.00 ± 0.03%, respectively; finally, for mathematics, the corresponding rates were 13.26 ± 0.01% and 1.18 ± 0.02%, respectively.

Overall, the average human and risky scoring rates were 13.64 ± 0.02% and 1.78 ± 0.01%, respectively, for all subjects, except for Japanese for second-grade elementary school students (13.27% and 10.99%, respectively), who provided particularly strict glyph shape evaluations, with a total of 937 513 answers. Note that the standard errors were small for all grades and subjects because the number of answers was large. Hereinafter, standard errors will be omitted.

For the 25 Japanese reading comprehension questions that required sentence answers, scoring thresholds based on similarity were scanned in 0.05 increments across the dataset. For PM, the risky and human scoring rates were 0% and 56%, respectively. Using similarity, the human scoring rate was reduced to 39% at the thresholds of 0.5 and 0.25, while the risky scoring rate remained lower than 2%. It is necessary to explore methods to reduce the human scoring rate for this type of question.

This experiment shows that by setting appropriate scoring methods and thresholds, the risky and human scoring rates can be suppressed for many questions, and the overall performance of the system almost reaches the target. This result guarantees the performance of the system equipped with an automatic scoring engine.

The results also revealed differences depending on the choice of scoring method. The characteristics of the answers indicate that it is better to establish a scoring method that is appropriate for each of them in advance.

The results provide an index for selecting a scoring method for each question. We also identified areas of weakness in each automatic scoring method and suggested directions for improvement.

### B. ANALYSIS OF DISAGREEMENT CASES BETWEEN HUMAN AND AUTOMATIC SCORING

Figs. 12 and 13 show various patterns of answers where incorrect scoring occurred. First, examples of risky scoring, *FP*, are presented in Fig. 12. *FPs* occurred in answers that used non-existent characters, as shown in Fig. 12(a) and (b). As only existing characters appear as recognition candidates, writing a non-existent kanji in a kanji vocabulary question may result in a correction using an existing character, thus leading to an *FP*. Hence, methods to determine whether written kanji components form a non-existent character need to be explored. Additionally, there are numerous instances where students' responses were nearly identical to the correct answer; however, these were nonetheless disqualified by the teacher, as the grading standards mandate clear and balanced handwriting, as illustrated in Fig. 12(c) and (d).

Moreover, *FN* cases may also arise, as shown in Fig. 13, because of variations in representation in instances with only one registered correct answer. Either all possible correct answers should be registered, or equivalence determination

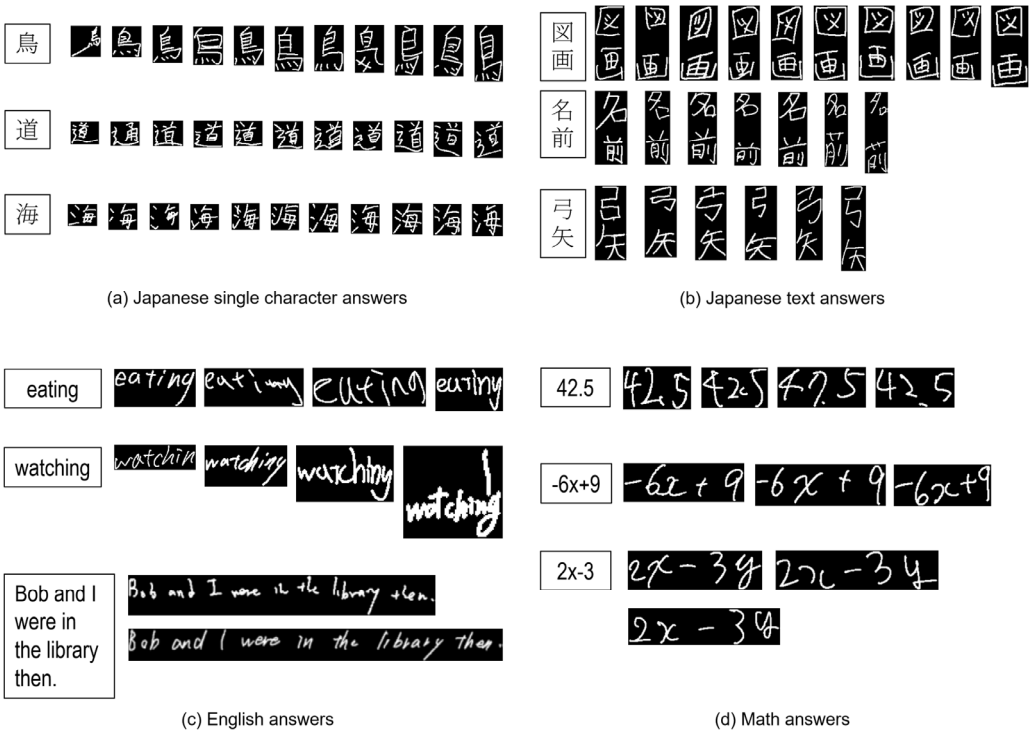



**FIGURE 12.** False positive examples that are incorrect answers judged as correct. The text boxes with white background contain exemplar answers while the black background boxes contain handwritten answers by students.

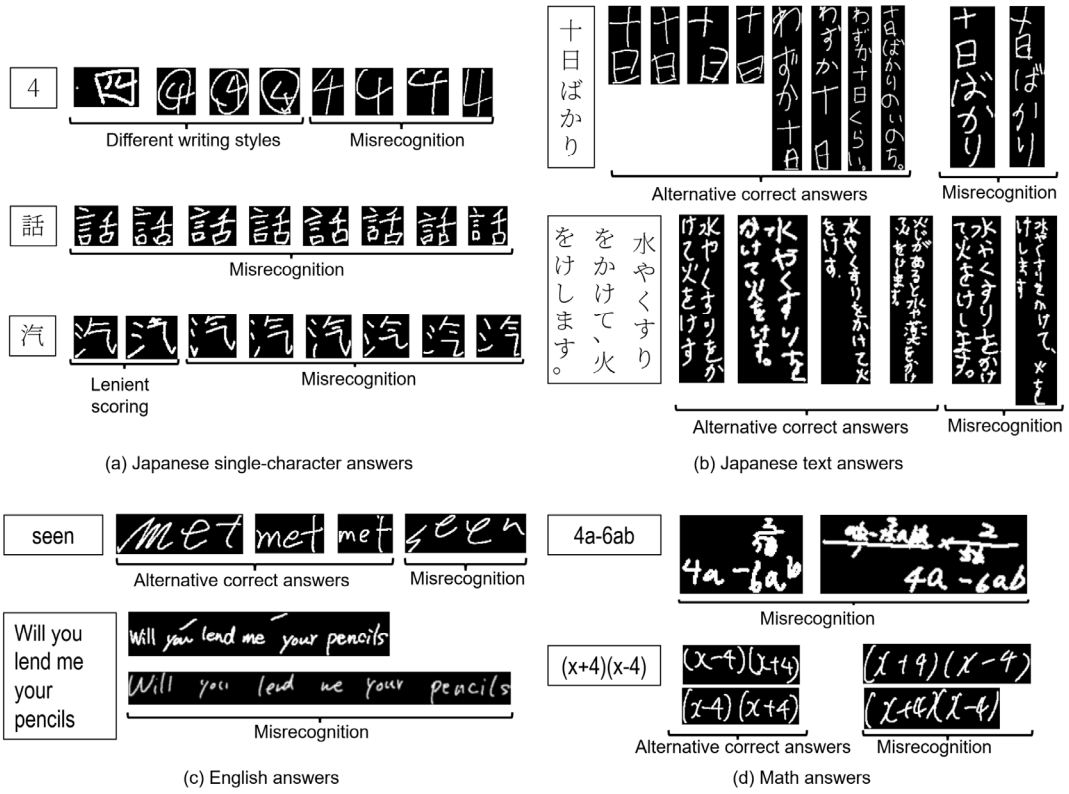



**FIGURE 13.** False negative examples that are correct answers judged as incorrect. The text boxes with white background contain exemplar answers while the black background boxes contain handwritten answers by students.

is required. Achieving the latter would likely necessitate the use of advanced recognition technology.

### C. EVALUATION OF UI

Subsequently, we evaluated the answerer and scorer UIs. For the answerer UI, we conducted a preliminary feasibility experiment involving 29 adult participants who were familiar with the use of computer terminals. In the experiment, a teacher familiar with the system provided instructions for taking a test using the system and distributed the questions. The participants used IoT Paper to write their answers, after which the system automatically scored and returned the results. Participants reviewed their scores (including the reported incorrect scores, which were then corrected by the scorers). Incorrect scores were reported via Google Forms, an external platform. Given the stable network environment, scoring results were returned within 1 min after test completion, allowing for prompt feedback.

Following the mock test, a questionnaire was distributed to all participants. Responses were received from 23 answerers and 3 scorers.

Respondents to the questionnaire ranged in age from 23 to 64 years, with an average of 43 years (19 males and 4 females). Answerers were asked to rate each of the following items on a 4-point scale; additionally, they were asked to provide open-ended responses: "How about the time it takes to return the scoring?" "How about the scoring results being returned immediately upon study?" "How about the system usability?" "How about the usability of reporting for incorrect scoring?" The response options for all these questions included "very satisfied," "slightly satisfied," "slightly unsatisfied," and "not satisfied at all."

Fig. 14 presents the answerers' survey results. The speed of score returns and expectations for learning effects were rated positively. However, some usability issues were noted, particularly concerning the UI design and the functionality for reporting incorrect scores.

To evaluate the scorer UI, three male scorers aged 28, 38, and 64 years responded to the same questionnaire. Regarding system usability, two scorers were slightly satisfied, and one was slightly unsatisfied. For the item "usability for correcting scores," one was very satisfied, and two were slightly satisfied. Although improvements are needed for scorer-facing UI components, the scoring correction function was evaluated positively.

### D. FUTURE ISSUES

First, we describe the issues identified during the evaluation of the automatic scoring performance and discuss the methods to address them.

Regarding Japanese for elementary school students, the risky scoring rate for single characters was approximately 5%, and it was particularly high in second grade, reaching almost 20%, as shown in Table 4. The risky scoring rate for multiple characters was high in the second grade, as shown in Table 5. This is attributed to the strict scoring criteria

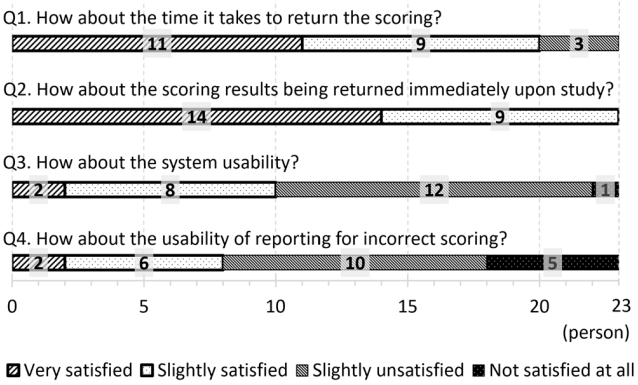



**FIGURE 14.** Survey results on the automatic scoring system from participants as answerers.

applied by certified language teachers. In early grades, Kanji instruction emphasizes character balance, stroke order, and stroke ending features. This problem is more pronounced for answers comprising a single or a few Kanji characters. The use of pattern similarity evaluation and calligraphic functions is considered to reduce the risky scoring rate.

For the 25 short-answer questions in Japanese reading comprehension, both perfect and similarity matching outcomes yielded high human scoring rates. Because it is necessary to assess the correctness of the content of the answers, the use of a language model is considered.

In English, the risky scoring rate reached approximately 3% for the 1st- and 2nd-year junior high school students, as listed in Table 6. Strict scoring criteria may be used for these grades. Pattern similarity evaluation may reduce the risky scoring rates.

Regarding mathematics, only third-grade junior high school students had a risky scoring rate of more than 2%, as shown in Table 7, which needs to be analyzed.

Although the overall human scoring rate remained below 30% and the risky scoring rate below 2%, certain individual questions still require targeted improvements. High scoring rates may discourage both scorers and answerers from using automatic scoring methods. We used default rubric settings in this experiment, but these can be refined by tailoring them for individual questions. It is challenging to refine or automate rubric settings to reflect scoring results.

Next, we describe the issues and countermeasures revealed by the UI evaluation. Neither the answerers nor the scorers were fully satisfied with the usability and reports of incorrect scoring. Regarding UI, the answerers were particularly dissatisfied with the pagination of the exam sheets. All operations other than writing during the test, including page feeding, were performed on the browser. It is necessary to design a UI that minimizes the difficulty of switching back and forth between writing and other operations. Our goal is to provide a stress-free UI that supports effective engagement with the learning cycle.

When the system is used for fully automated scoring, it expects the answerers to claim scoring errors that are to

be corrected by the scorer. This may increase the interaction between the answerers and the scorer, and change the paradigm of exam scoring. This must be evaluated in practice from instructional designs and learning theories.

In future studies, the scope of automatic scoring should be expanded. We plan to consider applying the language model to questions with line-drawing answers that were excluded from this study, as well as to long written answers.

## VII. CONCLUSION

In this paper, we proposed a system for both automatic scoring and scoring support for answers handwritten with an electronic pen to questions displayed on electronic paper. We also implemented the system as a web application. We digitized study drills in Japanese, English, and mathematics for all six grades of elementary school and three grades of junior high school; collected handwritten answers in Japanese, English, and mathematics from more than 50 students in each grade, and conducted automatic scoring experiments.

In these experiments, we used DNN-based handwriting recognition and applied automatic scoring based on either PM or TbS. By selecting appropriate scoring methods and setting thresholds using a validation set, we achieved average human and risky scoring rates below the target levels of 30% and 2%, respectively. Feasibility experiments on the implemented system showed that all the functions, question distribution, handwritten answering, scoring, return of scoring results, confirmation, reporting, and correction by the scorer operated smoothly. These results indicate a high level of satisfaction with the system's goals of immediate return of scoring results, and feasible UIs, although some functions and features must be improved.

This was the first milestone in the practical application of automatic scoring, and it provides the foundation for a cycle of collecting answer data and improving the accuracy of automatic scoring. However, some issues still remain. The risky scoring rates should be reduced for Japanese and English questions when precise writing is required. Additionally, rubrics could be refined for each question or each type of questions to further reduce human and risky scoring rates. Rubrics, reflecting performance outcomes, may be automatically refined.

In future work, we will extend scoring capabilities to include line-drawing and long descriptive answers, where language models may be especially useful. We also plan to enhance rubric flexibility and conduct demonstration experiments to evaluate system practicality, gather field-based data, and identify remaining challenges. Through these efforts, we aim to refine the system further and contribute to progress in automatic scoring research.

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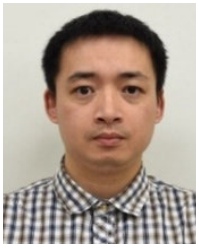
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